

2.4 DNS

The Internet's Directory Service



Kurose & Ross, 8th Ed.



How Do We Identify a Host?

Section: Introduction

Hostname

www.google.com
gaia.cs.umass.edu
www.amazon.com

Human-friendly,
mnemonic, easy to remember.



DNS
translates

IP Address

142.250.185.36
172.217.0.46
192.0.2.1

Fixed 4-byte, hierarchical,
preferred by routers.

Humans prefer hostnames · Routers prefer IP addresses · DNS bridges the gap

DNS — What Is It?

Section: Introduction

DNS = Domain Name System

① Distributed Database

A hierarchy of DNS servers distributed around the world — no single server holds all mappings.

② Application-Layer Protocol

Allows hosts to query the distributed database.
Runs over UDP · Port 53

DNS is not an application users interact with directly — it is invisible core infrastructure.

Service 1: Hostname-to-IP Translation

Section: Services Provided by DNS

What happens when you visit www.someschool.edu/index.html?

1

Browser extracts hostname

`www.someschool.edu` from the URL

2

Passes it to DNS client

The DNS client runs on the same machine as the browser

3

DNS client sends a query

A query message containing the hostname is sent to a DNS server

4

DNS client receives a reply

The reply includes the IP address for the hostname

5

TCP connection established

Browser connects to port 80 at that IP address and sends HTTP request

Services 2 & 3: Aliasing and Load Distribution

Section: Services Provided by DNS

Host Aliasing

A host can have a canonical hostname and one or more simpler aliases.

Canonical: relay1.west-coast.enterprise.com

Aliases: enterprise.com · www.enterprise.com

DNS returns the canonical name and its IP address.

Mail Server Aliasing

Email addresses must be simple (e.g., bob@yahoo.com) even if the real mail server has a complex hostname.

DNS resolves this via MX records.

Bonus: a company can use the same alias for both web and mail servers.

Load Distribution

Popular sites (cnn.com) are replicated across many servers, each with a different IP.

DNS stores the full set of IPs. With each reply, DNS rotates the ordering → clients use the first IP → traffic is distributed evenly.

Why Not a Single Centralized DNS Server?

Section: Why Not Centralized?

Single Point of Failure

If the server goes down, the entire Internet loses DNS resolution.

Geographic Distance

One server in the US creates huge latency for users in Asia, Africa, etc.

Traffic Volume

Billions of queries per day — impossible to handle on one machine.

Maintenance

Every new host or address change must update the same monolithic database.

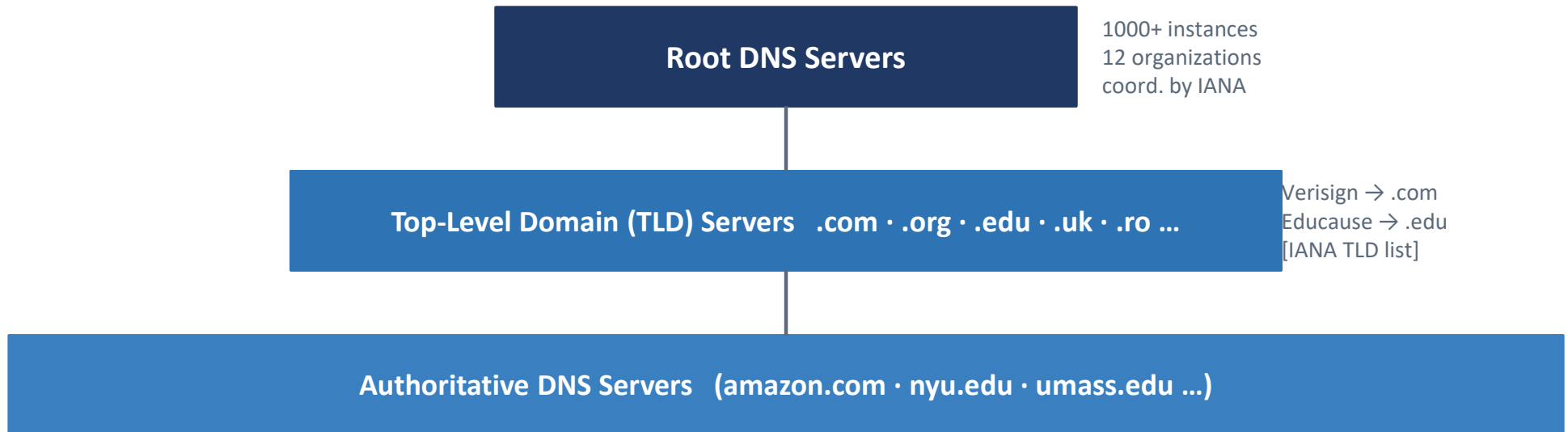
Massive Database

Must store and constantly update mappings for ALL hosts on the Internet.

→ DNS is distributed by design. No single server has all the mappings.

The DNS Hierarchy: Three Classes of Servers

Section: Distributed Hierarchical Database



+ Local DNS Servers (per ISP): act as proxies — not in the hierarchy but central to resolution

Local DNS Server (Default Name Server per ISP)
Provided to host via DHCP · Acts as proxy into the hierarchy

DNS Resolution: Step-by-Step (8 Messages)

Section: How DNS Works

Example: `cse.nyu.edu` → `gaia.cs.umass.edu`

- ① `cse.nyu.edu` → `dns.nyu.edu`
"What is the IP of `gaia.cs.umass.edu`?" (recursive query)
- ② `dns.nyu.edu` → Root Server
"Who handles `.edu`?"
- ③ Root → `dns.nyu.edu`
"TLD servers for `.edu` are at: [IP list]"
- ④ `dns.nyu.edu` → TLD `.edu` server
"Who handles `umass.edu`?"
- ⑤ TLD → `dns.nyu.edu`
"Authoritative for `umass.edu` is `dns.umass.edu`"
- ⑥ `dns.nyu.edu` → `dns.umass.edu`
"What is the IP of `gaia.cs.umass.edu`?"
- ⑦ `dns.umass.edu` → `dns.nyu.edu`
"IP = 128.119.40.0"
- ⑧ `dns.nyu.edu` → `cse.nyu.edu`
"IP = 128.119.40.0" ← final answer

Iterative vs. Recursive Queries

Section: How DNS Works

Recursive Query

The client asks the server to fully resolve the query on its behalf.

Example: the requesting host sends its query to the local DNS server and waits for the final IP address. The local server does all the work.

Typical use: requesting host → local DNS server

Iterative Query

The server responds with a referral (not the final answer). The original requester follows up with the next server.

Example: local DNS server asks root → TLD → authoritative, each returning the next server to contact.

Typical use: local DNS server → root, TLD, authoritative

In practice: host → local DNS is recursive · local DNS → root/TLD/authoritative is iterative

DNS Caching

Section: DNS Caching

When a DNS server receives a reply, it caches the mapping in local memory for future queries.

Reduced Latency

Cached replies are returned immediately without traversing the hierarchy.

Fewer Messages

Root servers are bypassed for the vast majority of queries.

TTL Controls Freshness

Time-To-Live (TTL) field: typically 2 days. Expired entries are discarded to stay consistent.

Example

apricot.nyu.edu

asks dns.nyu.edu for cnn.com → resolved & cached

Later that day:

kiwi.nyu.edu asks for cnn.com → dns.nyu.edu replies instantly from cache

After 2 days:

TTL expires → dns.nyu.edu discards the cached mapping

DNS Resource Records (RRs): Types A and NS

Section: DNS Records and Messages

Each RR is a tuple: (Name, Value, Type, TTL)

Type A

Name = hostname

Value = IP address of the hostname

(relay1.bar.foo.com, 145.37.93.126, A)

Type NS

Name = domain (e.g., foo.com)

Value = hostname of the authoritative DNS server for that domain

(foo.com, dns.foo.com, NS)

Important rule:

- If a server IS authoritative for a hostname → it holds a Type A record.
- If a server is NOT authoritative → it holds a Type NS record (pointer to the right server) + a Type A record giving the IP of that DNS server.

Resource Records: Types CNAME and MX

Section: DNS Records and Messages

Type CNAME

Name = alias hostname

Value = canonical hostname for the alias

(foo.com, relay1.bar.foo.com, CNAME)

Type MX

Name = alias hostname (e.g., foo.com)

Value = canonical name of the mail server

(foo.com, mail.bar.foo.com, MX)

Using CNAME and MX Together

- A company can use the same alias (e.g., enterprise.com) for both its web server and its mail server.
- Web server query → DNS client requests the CNAME record → gets relay1.west-coast.enterprise.com
- Mail server query → DNS client requests the MX record → gets mail1.west-coast.enterprise.com

DNS Messages

Section: DNS Records and Messages

DNS has exactly two message types: Query and Reply — both use the same format.

Header (12 bytes)

Identifier (16-bit) · query/reply flag · authoritative flag · recursion-desired · recursion-available · 4 count fields

Question Section

Hostname being queried · Query type (A, NS, MX ...)

Answer Section

RR(s) that directly answer the query

Authority Section

RR(s) for authoritative servers

Additional Section

Helpful extra RRs (e.g., Type A for a server named in an NS record)

Try it yourself: `nslookup <hostname>` — available on all Unix/Windows systems

Inserting Records into DNS

Section: Inserting Records

Scenario: You just founded Network Utopia and registered networkutopia.com

1

Register the domain

Contact an ICANN-accredited registrar (e.g., GoDaddy). The registrar verifies uniqueness and adds the domain to DNS.

2

Provide authoritative servers

Give the registrar names & IPs: dns1.networkutopia.com / 212.212.212.1 and dns2.networkutopia.com / 212.212.212.2

3

Registrar inserts into TLD .com

(networkutopia.com, dns1.networkutopia.com, NS)
(dns1.networkutopia.com, 212.212.212.1, A)

4

Configure your own server

Add Type A for www.networkutopia.com and Type MX for mail.networkutopia.com

DNS Vulnerabilities

Section: DNS Vulnerabilities

DDoS Attack on Root Servers

Flood root DNS servers with traffic → legitimate queries go unanswered.

Real event: October 21, 2002 — botnet sent ICMP pings to all 13 root IPs.

Defense: packet filtering + caching saved the day.

Man-in-the-Middle (MITM)

Attacker intercepts DNS queries and returns false IP addresses → users are redirected to malicious websites without knowing.

DNS Cache Poisoning

Attacker injects false records into a DNS server's cache → all clients querying that server receive wrong answers until TTL expires.

DNSSEC (DNS Security Extensions) — a secured version of DNS, gaining adoption across the Internet

Summary

2.4 DNS — The Internet's Directory Service

- DNS translates human-friendly hostnames to IP addresses — the core Internet directory service.
- DNS provides: hostname-to-IP, host aliasing, mail server aliasing, and load distribution.
- Architecture: hierarchical & distributed — Root → TLD → Authoritative → Local DNS servers.
- Resolution uses iterative & recursive queries; a single lookup can involve 8–10 DNS messages.
- Caching dramatically reduces latency and traffic; TTL controls how long entries remain valid.
- Resource Records: Type A (hostname→IP), NS (domain→auth server), CNAME (alias), MX (mail).
- New domains registered via ICANN-accredited registrars; records inserted into TLD servers.
- DNS can be attacked (DDoS, cache poisoning, MITM); DNSSEC provides cryptographic defense.